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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE

EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCTXL100 – TEXTILE SCIENCE

(2024 ADMISION ONWARDS)

Part A

Multiple Choice Questions

Answer All Questions

Each Question carries 1 mark

1. **List out the incorrect one of preparing acetate rayon.**
   b). oxidisation
2. **Which weave creates small localised dots or figures with the help of extra warp/weft yarns?**
   d). Spot Weave
3. **Select the process, which is not involved in the production of cotton yarn.**
   d). retting
4. **Which of the following is a common use of the net fabric?**
   d). Bridal Veils
5. **Polyester is known for its resistance to which of the following?**
   d). all of the above
6. **Find out the method of retting which is commonly used for flax processing.**
   a). water retting
7. **Which of the following is a characteristic of filament fiber?**
   b). long and continuous
8. **Identify one characteristic feature of fabric made with satin weave**.
   a). Glossy surface
9. **Which is the first synthetic fibre?**
   a). Nylon
10. **Select any one correct acid and alcohol for the preparation of polyester**
    a). Ethyleneglycol and Terephthalic acid

Part B

Fill in the Blanks Questions

Answer All Questions

Each Question carries 1 mark

11. The fiber extracted from the flax plant is known as **Linen.**
12. The direction of twist in yarn can be either **S or Z Twist.**
13. **Woolen** fabric are good insulators of heat.
14. **Plain weave** is the simplest type of weave.
15. Acrylic is non for its **UV resistance** to sunlight, which allows it to maintain colour and strength in outdoor use.
16. Unlike natural fibers, synthetic fibers are not very **absorbent**, which makes them less breathable.
17. The Distaff hold the fibers, which are then **twist** onto the spindle.
18. Before spinning, cocoons are boiled to soften the **sericin**, a protein that holds the fibers together.
19. A double cloth weave uses **Five** (number) of yarns to construct the fabric.
20. Cotton fibres have a **Kidney -shaped** cross-sectional shape

Part C

Answer in one Word or One Sentence

Answer All Questions

Each Question carries 1 mark

21. Cite any one type of felt.
Wool/needle
22. What is Slub?
Thickened areas of a fiber or yarn
23. Name one type of Novelty yarn?
Boucle, Chenille, Spot, Ratine (any one)
24. What happens to silk fibres when exposed to water?
Weaken fibres
25. What is the elasticity of nylon?
Very good elasticity, to return its original strength after stretching.
26. List out the feeling of wool fibre.
Soft, Warm, Bulkier texture.
27. How did acrylic polymer is made?
Made from polymer called polyacrylonitrile.
28. What Process aligns fibers parallel?
Drawing or Drafting/ Combing.

29. How does acetate rayon perform in terms of absorbency?

Low absorbency

30. Write in a sentence, the Odor emitted by burning acrylic fibers?

Chemical or meat smell/ melting plastic.

Part D

Short Answer Questions

Answer any 5 Questions

Each Question carries 2 marks

31. Differentiate between left hand twill and right-hand twill.

L.H.T Starts - upper left hand. Ends- lower right.

R.H.T Starts-upper right-hand Ends -lower left

32. What are the characterstics features of flax under microscopy?

Longitudinal view: flat and untwisted and swelled up portions and intervals that resemble the node of bamboo stick/node like structure.

Cross-section: cell wall appears thick and polygonal in shape.

33. What are the main advantages of Rotary Spinning?

High production speed, Consistent yarn Quality, Low energy consumption.

34. Explain the significance of the shining surface observed in discuss rayon fibres under a microscope.

Fibres smooth texture: Rayon is a man-made regenerated cellulose fibre, produced by extruding viscose through spinnerets, this process creates smooth, cylindrical filaments with very few irregularities.

Natural Lustre: light reflects easily off the smooth fibre surface, rayon looks silky and glossy, which is one of the reasons rayon is used as an artificial silk.

35. Explain pile weave and give two examples of fabrics created using this weave?

Pile - Soft projecting hair fibres, it produces a three-dimensional effect on the fabric.

eg: - Uncut pile - Terry Towel, Cut pile – Velvet**.**

36. **Differentiate between Blends and Mixtures**.

Blending is the process of combining two or more different types of fibres -Quality not considered. Same properties of fibres are used for blending.

A blend occurs when two or more different fibers are spun together into a single yarn before weaving or knitting. The fibers are mixed at the fiber stage, creating a yarn where the fibers are intimately combined.

Mixtures – same fibers of different properties- quality is considered. Different properties of fibres are used for mixing. A mixture refers to the use of different types of yarns made from different fibers in the warp and weft directions of woven fabrics or in different sections of knitted fabrics. Unlike blends, fibers in mixtures remain separate and are combined at the fabric construction stage.

Part E

Long Answer Questions

Answer any 2 Question

Each Question carries 5 marks

37.Discuss how handmade lace differs from machine made lace.

Handmade - Bobbin lace, Needle lace, tatting lace, crocheted lace
Machine made - Leavers Lace, Raschel Lace, Ratine Lace, Schiffli Lace

38.Explain the essential properties required for a successful textile fibre?

Strength (Tenacity)- The fiber must have sufficient tensile strength to withstand the stresses of spinning, weaving, and general wear and tear.
flexibility- A fiber must be able to bend without breaking to be manipulated and twisted around other fibers during yarn formation.

Uniformity- For consistent quality, fibers should have uniformity in length, diameter, and other properties.

Fibre length to width ratio- Fibers need to be significantly longer than they are wide to be spun into strong, smooth yarns.\

Cohesiveness or spinning quality- This is the ability of fibers to cling to one another, which is crucial for spinning

39.Write in detail on properties of viscose rayon.

Strength - Viscose has moderate dry strength, but it loses 40-70% of its strength when wet. **Elasticity**- Lacks elasticity; it has poor resilience., **Drapability**- Exhibits excellent drapability, which makes it suitable for flowing garments.

Heat adsorbent- Rayon begins to decompose at high temperatures (~150–200°C).Scorches easily and should be ironed at low temperatures.

Washability- Requires gentle handling during washing.

Shrinkage- Viscose rayon has a high tendency to shrink, especially when it becomes wet.

Reaction to acids and alkalis- **Weak acids**: fairly resistant. **Strong acids**: cause hydrolysis and weaken fiber. **Alkalis**: Strong alkalis (like caustic soda) damage viscose rayon; mild alkalis cause swelling.

Mildew- Viscose rayon is highly prone to mildew and fungal attack.

Dyeability- Viscose rayon has very high dyeability due to Its high moisture regain.

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40.Write in detail about a) Single yarn b) Cord yarn c) Ply yarn

Single yarn: twisting staple fibres.

Ply yarn: twisting two or more single yarn.

Cord yarn: Twisting two more ply yarns.

Prepared By

Ms.Sruthy Mohan

Asst.Professor

YMBC,Kothamangalam